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GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC: ELECTRIC STEERING COLUMN LOCK (ESCL) [G1551457]

DESCRIPTION AND OPERATION

ELECTRIC STEERING COLUMN LOCK CONTROL MODULE (ESCL)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If a control module or a component is out of specification and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter that has the resolution ability to view 3 decimal places. For example, on the 2 volts range can measure 1mV or 2 K Ohm range can measure 1 Ohm. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed through the 'diagnostic trouble code monitor' tab on the Jaguar Land Rover approved diagnostic equipment
- Check the JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and complete the recommendations as required.

The table below lists all diagnostic trouble codes (DTCs) that could be set in the electric steering column lock control module, for additional diagnosis and testing information refer to the relevant diagnosis and testing section. For additional information, refer to: Anti-Theft - Active (419-01A Anti-Theft - Active, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B100D-16	Column Lock Authorisation - Circuit voltage below threshold	▪ Electric steering column lock control module power supply too low ▪ Electric Steering Column Lock circuit short to ground, high resistance, open circuit ▪ Electric steering column lock control module internal failure	▪ Refer to the electrical circuit diagrams and check the electric steering column lock control module power and ground circuits for open circuit, high resistance ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs. Perform routine - On Demand Self Test (0202). If the fault persists, install a new electric steering column lock control module
B100D-29	Column Lock Authorisation - Signal invalid	▪ Electric steering column lock control module internal sensor signal invalid	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs. Perform routine - On Demand Self Test (0202). If the fault persists, install a new electric steering column lock control module
B100D-42	Column Lock Authorisation - General memory failure	▪ Electric steering column lock control module internal failure	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs. Perform routine - On Demand Self Test (0202). Perform a CAN network integrity test. If the fault persists, install a new electric steering column lock control module
B100D-51	Column Lock Authorisation - Not programed	▪ Electric Steering Column Lock not programed	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, re-configure the electric steering column lock control module as a new module
B100D-62	Column Lock Authorisation - Signal compare failure	▪ Mismatch between electric steering column lock and central junction box security data ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and perform routine - On Demand Self Test (0202). If the fault persists, re-configure as a new module ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B100D-64	Column Lock Authorisation - Signal plausibility failure	▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Anti-lock brake system fault ▪ Engine system fault ▪ Body system fault	**NOTE:** Prior to clearing this DTC, perform the Vehicle Functional Reset routine ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check the engine control module for related DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check the central junction box for related DTCs and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B100D-72	Column Lock Authorisation - Actuator stuck open	▪ Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock control module internal failure	▪ Turn the steering wheel to make sure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs. Perform routine - On Demand Self Test (0202). If the fault persists, install a new electric steering column lock control module
B100D-73	Column Lock Authorisation - Actuator stuck closed	▪ Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock control module internal failure	▪ Turn the steering wheel to make sure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs. Perform routine - On Demand Self Test (0202). If the fault persists, install a new electric steering column lock control module
B100D-77	Column Lock Authorisation - Commanded position not reachable	▪ Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock control module internal failure	▪ Turn the steering wheel to make sure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs. Perform routine - On Demand Self Test (0202). If the fault persists, install a new electric steering column lock control module
B100D-92	Column Lock Authorisation - Performance or incorrect operation	▪ Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock control module internal failure	▪ Turn the steering wheel to make sure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs. Perform routine - On Demand Self Test(0202). If the fault persists, install a new electric steering column lock control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B100D-94	Column Lock Authorisation - Unexpected operation	▪ Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock control module internal failure	▪ Turn the steering wheel to make sure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs. Perform routine - On Demand Self Test (0202). If the fault persists, install a new electric steering column lock control module
B100D-96	Column Lock Authorisation - Component internal failure	▪ Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock control module internal failure	**NOTE:** Prior to clearing this DTC, perform the Vehicle Functional Reset routine ▪ Turn the steering wheel to make sure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs. Perform routine - On Demand Self Test (0202). If the fault persists, install a new electric steering column lock control module
U0001-88	High Speed CAN Communication Bus - Bus off	▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U3002-81	Vehicle Identification Number - Invalid serial data received	▪ Invalid data received by the electric steering column lock control module ▪ Electric steering column lock control module is not configured correctly	**NOTE:** This DTC is for event information and does not indicate a fault ▪ No action necessary. Ignore this DTC ▪ Using the Jaguar Land Rover approved diagnostic equipment, re-configure the electric steering column lock control module with the latest level software

cmVlX2FtZzsyMDI1LTEyLTE5VDA2OjU1OjQ1LjU1OFo7MjA3LjE4MS4zLjEzMTtTQUpERDFHWDRMQ0s2NzQ5Nw==